

Understanding CIFAR-10 & CNN Fundamentals

A Comprehensive Technical Reference Guide | Image Tensors, Channel Progression & Feature Maps

1. Bit Depth & Color Representation

Understanding how computers store color information forms the foundation of computer vision.

8-Bit Color System (3-3-2 RGB)

Uses 1 byte (8 bits) per pixel to represent 256 colors.
Allocation follows human vision sensitivity:

- **Red:** 3 bits = 8 shades
- **Green:** 3 bits = 8 shades
- **Blue:** 2 bits = 4 shades (human eyes are less sensitive to blue)

Truecolor / 24-Bit (CIFAR-10 Standard)

Uses 3 full bytes per pixel (1 byte per channel):

- **Red Channel:** 8 bits = 256 intensity levels (0–255)
- **Green Channel:** 8 bits = 256 intensity levels (0–255)
- **Blue Channel:** 8 bits = 256 intensity levels (0–255)
- **Total Palette:** 16.7 Million unique colors

2. CIFAR-10 Dataset Architecture

The CIFAR-10 dataset consists of 60,000 real-world color images categorized into 10 mutually exclusive classes.

Property	Specification	Description
Spatial Resolution	32 × 32 pixels	Total grid squares = 1,024 spatial locations per channel.
Color Channels	3 (RGB)	Three stacked 2D grids (Red, Green, Blue).
Value Scale	0 to 255	Integer intensity values (0 = absolute dark, 255 = maximum brightness).
Total Values per Image	3,072 numbers	Calculated as $32 \times 32 \times 3 = 3,072$ scalar numbers.
Dataset Distribution	60,000 Total Images	50,000 training images and 10,000 testing images.

3. Deep Learning Framework Tensor Shapes

Frameworks organize multi-dimensional arrays (tensors) in different axis orders in memory:

Framework	Layout Format	Single Image Shape	Batch Shape (N Images)
TensorFlow / Keras / OpenCV	HWC (Height, Width, Channels)	(32, 32, 3)	(N, 32, 32, 3)
PyTorch	CHW (Channels, Height, Width)	(3, 32, 32)	(N, 3, 32, 32)

Key Takeaway: What a Model Sees

When a neural network receives a CIFAR-10 image, it does not perceive visual objects. It ingests an ordered array of

4. Convolutional Layer Mechanics: Conv1 (3 → 32 Channels)

When defining a layer such as `Conv2d(in_channels=3, out_channels=32)`, the network transforms raw pixel intensities into feature representations.

Input Space (3 Channels)

Starts with raw pixel light intensities: 3 channels corresponding to Red, Green, and Blue.

32 Learned Filters

Applies 32 distinct 3D kernels. Each filter scans all 3 channels simultaneously to detect specific patterns.

What Do the 32 Output Feature Maps Represent?

Instead of inspecting raw colors, the model now evaluates **32 abstract visual features** across the image:

- Filter 1 highlights **horizontal edges**.
- Filter 2 detects **vertical boundaries**.
- Filters 3–10 highlight **diagonal lines, corners, and texture gradients**.
- Filters 11–32 detect **color contrasts and specific color transitions**.

5. Spatial Execution: Patch vs. Parallel Processing

Perspective	Unit of Operation	How It Works
Receptive Field (Spatial)	3×3 Local Patch	A filter processes a small 3×3 neighborhood across all 3 input depth channels at a time ($3 \times 3 \times 3 = 27$ weights).
Hardware Execution (GPU)	Entire Image at Once	GPUs do not use slow loops to slide pixel-by-pixel. Convolution is converted to parallel matrix multiplications, calculating all output values simultaneously.

6. Core Rule: Number of Feature Maps

Number of Output Feature Maps = Number of Filters Applied

- 1 Filter applied to input = 1 2D Feature Map output.
- 32 Filters applied to input = 32 stacked 2D Feature Maps output.

7. Summary Workflow

1. **Raw Image Input:** Shape `(3, 32, 32)` — 3 color channels.
2. **First Convolution (Conv1):** Applies 32 filters of size $3 \times 3 \times 3$.
3. **Output Feature Tensor:** Shape `(32, 32, 32)` — 32 feature maps highlighting edges and shapes.
4. **Deeper Layers (Conv2, Conv3):** Combine simple features (edges) into complex structures (wheels, eyes, wings).